Safe Multi-Agent Reinforcement Learning with Model-Based World Models

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Abstract

Deploying multi-agent reinforcement learning (MARL) in safety-critical applications requires both strict safety guarantees and sample efficiency. Existing constrained policy optimization methods are inherently *reactive*, requiring agents to experience real-world failures to learn safety, while model-based approaches often ignore safety constraints entirely. We propose *Safe Dreamers*, the first framework unifying multi-agent world models with Lagrangian safety constraints. By learning to predict costs in a latent space, Safe Dreamers enables agents to be *proactive*: they learn to avoid safety violations in imagination, removing the need to experience them in reality. Evaluated on the StarCraft Multi-Agent Challenge (SMAC) with strict casualty constraints, Safe Dreamers achieves competitive task performance with *25× greater sample efficiency* than baselines, successfully satisfying constraints where reactive methods fail.

1 Introduction

Deploying multi-agent reinforcement learning (MARL) in real-world applications requires satisfying two often conflicting objectives: maximizing task performance and strictly adhering to safety constraints. In domains such as autonomous warehouse logistics, coordinated drone fleets, and power grid management, agents must learn complex cooperative behaviors without causing catastrophic failures during the learning process. However, standard RL paradigms rely on trial-and-error, which poses unacceptable risks in safety-critical environments where a single collision or violation can be prohibitively expensive or dangerous.

Current approaches to *Safe Reinforcement Learning (Safe RL)*—defined as learning policies that maximize reward while satisfying a set of cost constraints—typically rely on constrained policy optimization methods, such as MACPO Gu et al. (2021). While effective at converging to safe policies eventually, these model-free methods suffer from severe sample inefficiency, often requiring millions of environment interactions to learn even simple tasks. Conversely, model-based MARL approaches, such as MAMBA Egorov and Shpilman (2022), have demonstrated remarkable sample efficiency by learning world models that simulate environment dynamics. Yet, these methods traditionally focus solely on reward maximization, ignoring safety constraints and leaving agents blind to the potential costs of their actions during planning.

This creates a critical gap: practitioners are forced to choose between the sample efficiency of model-based methods and the safety guarantees of constrained policy optimization. To bridge this gap, we propose that the key to safe and efficient MARL lies in *imagination*: if agents can accurately predict the safety consequences of their actions within a learned world model, they can learn to avoid unsafe behaviors without ever executing them in the real world.

We introduce *Safe Dreamers*, the first framework that integrates multi-agent world models with Lagrangian safety constraints. Our approach extends the world model to jointly learn the dynamics of both rewards and costs in a latent space, enabling agents to "dream" potential futures and evaluate their safety during training. By optimizing policies within this constrained imagination, Safe Dreamers learns to satisfy safety constraints proactively rather than reactively.

Our contributions are as follows:

- We propose *Safe Dreamers*, a novel Model-Based Safe MARL framework that unifies world model learning with Lagrangian constraint optimization.
- We introduce a *Cost-Aware World Model* architecture that predicts safety costs alongside system dynamics, enabling interpretable safety planning in latent space.
- We empirically demonstrate that Safe Dreamers achieves *25× greater sample efficiency* than state-of-the-art model-free baselines (MACPO) on the StarCraft Multi-Agent Challenge (SMAC).
- We show that our method successfully satisfies strict safety constraints (e.g., zero friendly fire or casualty limits) in scenarios where baselines fail to find any feasible solution.

2 Related Work

The fundamental limitation of current Safe RL is its reliance on real-world failure. Traditional approaches formulate safety as a Constrained MDP Altman (1999) and enforce constraints via Lagrangian relaxation Liu et al. (2020) or trust-region updates Achiam et al. (2017). In the multi-agent domain, methods like MACPO Gu et al. (2021) successfully extend these disparate constraints to cooperative settings. However, theoretical guarantees aside, these methods are inherently *reactive*: they require the agent to sample unsafe trajectories in the environment to estimate the cost gradients. In practice, this means agents must "learn by dying"—accumulating thousands of safety violations to map out the dangerous regions of the state space. For safety-critical systems like drone swarms or power grids, this retroactive learning is unacceptable; safety must be ensured *before* the violation occurs, not learned from it.

Model-based reinforcement learning (MBRL) offers a path to reduce real-world interaction by substituting it with planning in a learned world model Hafner et al. (2020); Krinner et al. (2025). In multi-agent domains, methods like MAMBA Egorov and Shpilman (2022) have successfully scaled this "imagination" to complex coordination tasks, achieving high sample efficiency. However, these frameworks have traditionally focused on a single objective: reward maximization. Without an explicit mechanism to represent and predict risk, the "imagination" of these agents remains incomplete. They can plan for victory but cannot plan for safety, leaving them vulnerable to the same catastrophic failures as standard RL, albeit with fewer samples. The missing piece is a world model that understands danger as well as it understands dynamics.

We propose that the solution lies in combining these paradigms to enable *proactive* safety. Recent single-agent efforts like SafeDreamer Huang et al. (2024) have demonstrated that world models can be augmented to predict costs, allowing agents to "feel" the penalty of a safety violation in imagination without executing it. While other multi-agent approaches have attempted distinct safety mechanisms such as shielding Xiao et al. (2023), they often rely on hand-crafted rules or simplified transition models. Safe Dreamers generalizes the proactive imagination of SafeDreamer to the multi-agent setting, providing the first fully learned, end-to-end framework where teams of agents learn to avoid danger by optimizing policies within a safe, imagined simulator.

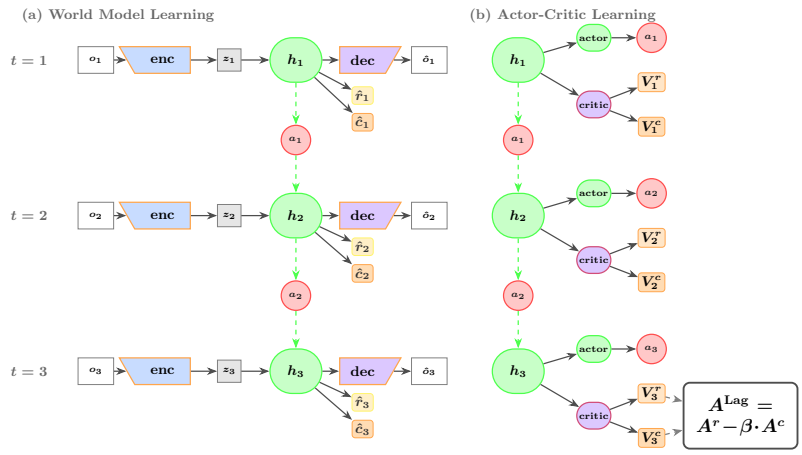


Figure 1: Training Phase: (a) **Learning the Safety Dynamics:** The world model learns to predict costs alongside rewards, embedding the safety landscape into the latent space. (b) **Proactive Safety via Imagination:** The actor optimizes a Lagrangian objective A^{Lag} entirely within imagination, effectively "violating" constraints in the simulator to learn safe behaviors without real-world risk.

3 Methodology

We present *Safe Dreamers*, a framework for safe multi-agent coordination that leverages world models to strictly enforce safety constraints. In this section, we first formally define the problem as a Constrained Dec-POMDP. We then detail our proactive safety mechanism via latent imagination and present the full algorithm and network architecture.

3.1 Preliminaries

We formulate the safe multi-agent coordination problem as a Constrained Decentralized Partially Observable Markov Decision Process (C-Dec-POMDP), defined by the tuple $\mathcal{M} = \langle \mathcal{I}, \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \mathcal{C}, \Omega, \mathcal{O}, \gamma, d \rangle$. Here, $\mathcal{I} = \{1, \ldots, N\}$ is the set of agents; $\mathcal{S}$ is the global state space; $\mathcal{A} = \times_{i \in \mathcal{I}} \mathcal{A}^i$ is the joint action space; and $\mathcal{O} = \times_{i \in \mathcal{I}} \mathcal{O}^i$ is the joint observation space. At each timestep t, each agent i observes a partial observation $o_t^i \in \mathcal{O}^i$ and selects an action $a_t^i \in \mathcal{A}^i$. The environment transitions to a new state $s_{t+1} \sim \mathcal{P}(\cdot | s_t, \mathbf{a}_t)$ based on the joint action $\mathbf{a}_t$, yielding a shared global reward $r_t = \mathcal{R}(s_t, \mathbf{a}_t)$ and a shared safety cost $c_t = \mathcal{C}(s_t, \mathbf{a}_t)$.

The objective is to find a joint policy π that maximizes the expected discounted return $J_R(\pi) = \mathbb{E}_\pi[\sum_{t=0}^{\infty} \gamma^t r_t]$ subject to a safety constraint on the expected discounted cost $J_C(\pi) = \mathbb{E}_\pi[\sum_{t=0}^{\infty} \gamma^t c_t] \leq d$, where d is a pre-defined safety threshold.

3.2 Proactive Safety via Latent Imagination

The Safe Dreamers framework is built on the core insight that safety is best learned through imagination, not experience. Instead of letting agents fail in the real world to update a safety critic (as in model-free methods), we first build a *safe simulator* in the latent space and then use it to train agents to avoid dangerous states proactively. This process operates in three iterative phases.

Phase 1: Learning the Safety Dynamics. The world model serves as the agent's internal simulator. Unlike standard MBRL, which focuses purely on dynamics and rewards, our world model must also approximate the "safety landscape" of the environment. We extend the MAMBA architecture to learn a joint distribution over observations, rewards,

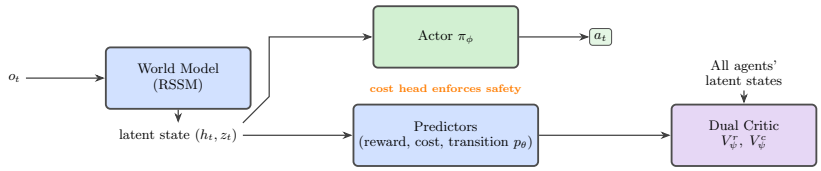


Figure 2: Method Overview: Integration of the World Model, Actor, and Safety Critics.

and safety costs. Following the RSSM framework, the model consists of six components:

$$\text{RSSM} = \begin{cases} \text{Recurrent model:} & h_t^i = f_\theta(h_{t-1}^i, s_{t-1}, a_{t-1}) \\ \text{Representation model:} & s_t^i \sim q_\theta(s_t^i | h_t^i, o_t^i) \\ \text{Transition predictor:} & \hat{s}_t^i \sim p_\theta(\hat{s}_t^i | h_t^i) \\ \text{Observation predictor:} & \hat{o}_t^i \sim p_\theta(\hat{o}_t^i | h_t^i, s_t^i) \\ \text{Reward predictor:} & \hat{r}_t^i \sim p_\theta(\hat{r}_t^i | h_t^i, s_t^i) \\ \text{Cost predictor:} & \hat{c}_t^i \sim p_\theta(\hat{c}_t^i | h_t^i, s_t^i) \end{cases}$$

where h_t^i is the deterministic hidden state, s_t^i is the stochastic latent state, and o_t^i, a_t^i, r_t^i, c_t^i are observations, actions, rewards, and costs for agent i. The world model is trained to minimize the joint objective $\mathcal{L}_{\text{model}}$:

$$\begin{aligned} \mathcal{L}_{\text{model}} &= \mathcal{L}_{\text{obs}} + \mathcal{L}_{\text{reward}} + \mathcal{L}_{\text{cost}} + \beta_{\text{KL}} \mathcal{L}_{\text{KL}} \\ &= -\sum_{t=1}^{T} \ln p_\theta(o_t^i | h_t^i, s_t^i) - \sum_{t=1}^{T} \ln p_\theta(\hat{r}_t^i | h_t^i, s_t^i) \\ &\quad - \sum_{t=1}^{T} \ln p_\theta(\hat{c}_t^i | h_t^i, s_t^i) \\ &\quad + \beta_{\text{KL}} \sum_{t=1}^{T} \text{KL}(q_\theta(s_t^i | h_t^i, o_t^i) \| p_\theta(\hat{s}_t^i | h_t^i)) \end{aligned}$$

where β_{KL} scales the regularization term. The KL divergence is computed per categorical dimension and then averaged across dimensions to form a scalar value per step. We additionally employ KL balancing to regularize the prior and posterior separately before the final aggregation for optimization. Crucially, adding the cost prediction loss $-\ln p_\theta(\hat{c}_t^i | \cdot)$ ensures that the learned latent state s_t encodes safety-critical information alongside task-relevant features. This allows the agent to predict not just the *outcome* of an action, but its *risk*, directly from the low-dimensional latent representation.

Phase 2: Proactive Safety via Imagination. Once the world model captures the safety dynamics, the agent uses it to generate rollouts for policy optimization. In this phase, we freeze the world model and train the actor and critic entirely within the latent imagination. The agent generates imagined trajectories of horizon $H = 15$ starting from states s_0 sampled from the replay buffer. At each step τ of the imagination, the actor $\pi_\phi(a_\tau | s_\tau)$ selects an action, and the world model predicts the next latent state $s_{\tau+1} \sim p_\theta(s_{\tau+1} | s_\tau, a_\tau)$, reward $\hat{r}_\tau$, and cost $\hat{c}_\tau$.

To evaluate the policy, we compute value targets using Generalized Advantage Estimation (GAE). Unlike standard Dreamer, we maintain *two* value functions: a reward value $V_\psi^r(s)$ and a safety cost value $V_\psi^c(s)$. The Lagrangian advantage A^{Lag} combines these signals:

$$A_\tau^{\text{Lag}} = A_\tau^r - \beta \cdot A_\tau^c \tag{1}$$

where A_τ^r and A_τ^c represent the advantage for reward and cost, respectively. By optimizing this joint advantage, the actor effectively "hallucinates" safety violations and learns to avoid them, converging to a constrained policy without exposing the real agent to danger. We update the policy using Proximal Policy Optimization (PPO) Schulman et al. (2017). Unlike purely analytic gradient methods (e.g., DreamerV2), applying PPO's clipped objective

Algorithm 1: Safe Dreamers: Safe Multi-Agent Model-Based RL

Input: Environment $\mathcal{E}$, cost limit d, horizon H
Initialize: World model θ, actor ϕ, dual critic ψ, $\beta = 0$, buffer $\mathcal{D}$
while *not converged* **do**
  // Collect data
  Collect episodes using policy π_ϕ, store in $\mathcal{D}$

  for *update step* **do**
    ▷ **Phase 1: Learning the Safety-Aware World Model**
    Sample batch $\{(o_t, a_t, r_t, c_t)\} \sim \mathcal{D}$
    Encode observations: $e_t \leftarrow \text{Encoder}_\theta(o_t)$
    Compute latent states via RSSM: $s_t \sim q_\theta(s_t | s_{t-1}, a_{t-1}, e_t)$
    Compute loss:
    $\mathcal{L}_{\text{model}} = \mathcal{L}_{\text{recon}} + \mathcal{L}_r(\hat{r}_\theta(s_t), r_t)$
    $+\mathcal{L}_c(\hat{c}_\theta(s_t), c_t) + \text{KL}(q_\theta \| p_\theta)$
    Update θ using $\nabla_\theta \mathcal{L}_{\text{model}}$

    ▷ **Phase 2: Proactive Safety via Imagination**
    Sample initial states: $s_0 \sim q_\theta$
    Roll out imagined trajectories for H steps:
    for $\tau = 0$ *to* H **do**
      $a_\tau \sim \pi_\phi(\cdot | s_\tau)$
      $s_{\tau+1} \sim p_\theta(\cdot | s_\tau, a_\tau)$
      $\hat{r}_\tau \leftarrow \hat{r}_\theta(s_\tau), \quad \hat{c}_\tau \leftarrow \hat{c}_\theta(s_\tau)$
    Compute returns: $R_\tau, C_\tau \leftarrow \text{TD-}\lambda(\hat{r}, \hat{c}, V_\psi)$
    Compute advantages:
    $A^r_\tau \leftarrow R_\tau - V^r_\psi(s_\tau), \quad A^c_\tau \leftarrow C_\tau - V^c_\psi(s_\tau)$
    $A^{\text{Lag}}_\tau \leftarrow A^r_\tau - \beta \cdot A^c_\tau$ (Lagrangian advantage)
    Update actor with PPO:
    $\rho_\tau \leftarrow \frac{\pi_\phi(a_\tau | s_\tau)}{\pi_{\phi_{\text{old}}}(a_\tau | s_\tau)}$
    $\mathcal{L}_{\text{actor}} \leftarrow -\mathbb{E}[\min(\rho_\tau A^{\text{Lag}}_\tau, \text{clip}(\rho_\tau, 1-\epsilon, 1+\epsilon) A^{\text{Lag}}_\tau)]$
    Update ϕ via gradient ascent
    Update dual critic:
    $\mathcal{L}_{\text{critic}} \leftarrow \mathbb{E}[(V^r_\psi - R_\tau)^2 + (V^c_\psi - C_\tau)^2]$
    Update ψ

    ▷ **Phase 3: Adaptive Lagrangian Update**
    $\bar{C} \leftarrow$ mean episode cost
    // Penalize violations seen in rollout
    $\beta \leftarrow \max(0, \beta + \alpha_\beta(\bar{C} - d))$

within imagination prevents the policy from exploiting transient errors in the learned safety dynamics, ensuring stable convergence in the constrained landscape.

Phase 3: Adaptive Lagrangian Update. To balance task performance and safety, we employ an adaptive Lagrangian multiplier β. This multiplier acts as a "safety thermostat": if the agent's recent behavior (measured by $\bar{C}$) exceeds the cost limit d, β increases, forcing the actor to prioritize safety in the next imagination phase. If the agent is safe, β decreases, allowing it to pursue higher rewards.

$$\beta \leftarrow \max(0, \beta + \alpha_\beta(\bar{C} - d)) \tag{2}$$

where α_β is the learning rate that controls the sensitivity of the Lagrangian update. By adjusting β based on real-world feedback, we ensure that any mismatch between the world model's cost predictions and reality is corrected. The complete training procedure is summarized in Algorithm 1.

3.3 Network Architecture

All networks use 256 hidden dimensions, as illustrated in Figure 2. The world model consists of: (1) encoder and decoder (2-layer MLPs), (2) RSSM with 256-dim deterministic state (GRU) and 1024-dim stochastic state (32 categorical distributions, 32 classes each), (3) 3-layer self-attention in the transition model (0.1 dropout), and (4) reward/cost predictors (2-layer MLPs). The actor is a 4-layer MLP outputting action logits. The dual critic uses single-layer self-attention followed by separate heads for V^r and V^c.

In the multi-agent setting, each agent has its own encoder and actor (decentralized execution), while the world model and critic are centralized. Coordination is achieved through self-attention in the RSSM transition (3 layers) and critic (1 layer).

3.4 Execution

During execution, each agent maintains its own copy of the world model to enable decentralized decision-making, consistent with the Centralized Training with Decentralized Execution (CTDE) paradigm. At each decision step, agents exchange latent states and action logits through a multi-head attention mechanism (8 heads with positional encoding), implementing all-to-all communication across agents. The attention block is applied before action selection, aggregating teammate information with computational complexity $\mathcal{O}(n^2 \cdot d)$ per step, where n is the number of agents and d is the feature dimension. Each agent then uses its observation and the attention-aggregated information to select actions via the trained actor policy.

While standard self-attention scales quadratically with the number of agents, this cost is negligible for the agent counts typical in SMAC (up to $n = 27$). For massive-scale scenarios (e.g., hundreds of agents), this communication block can be seamlessly replaced with more scalable alternatives without altering the core Safe Dreamers framework. Valid options include Mean Field approximations **?** which reduce complexity to $\mathcal{O}(1)$ by averaging neighbor embeddings, or sparse attention mechanisms **?** that limit interaction to the k-nearest neighbors for $\mathcal{O}(n \cdot k)$ scaling. In our evaluated domain, however, the all-to-all attention provides the necessary expressivity to capture complex coordination patterns without hitting computational bottlenecks.

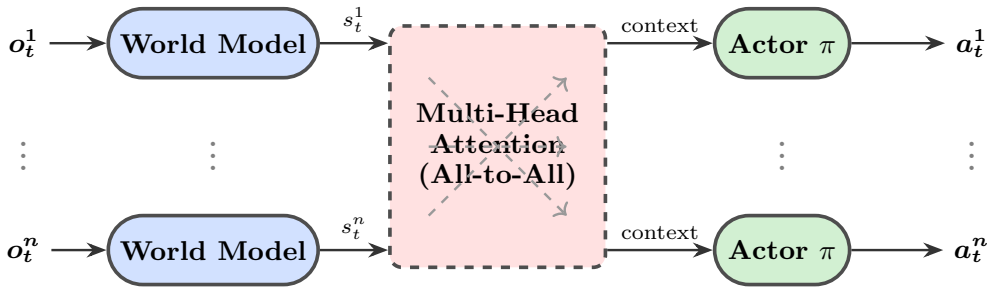


Figure 3: Execution Phase: Each agent maintains its own world model copy. At each decision step, all agents communicate via multi-head attention (all-to-all, $\mathcal{O}(n^2 \cdot d)$ complexity) before action selection, enabling decentralized execution.

4 Experiments

Our experiments are designed to evaluate the hypothesis that learning safety in latent imagination is superior to learning it from real-world failures. Specifically, we address three core research questions:

- **RQ1 (Proactive Safety):** Can Safe Dreamers maintain high task performance while strictly adhering to safety constraints, whereas reactive baselines fail?

- **RQ2 (Sample Efficiency):** Does learning in imagination quantifyably reduce the real-world samples needed to converge to a safe policy?

- **RQ3 (Mechanism):** Are the world model's cost predictions and the Lagrangian mechanism essential for this success? (addressed in Section 4.4).

4.1 Experimental Setup

Domain: StarCraft II (SMAC). We evaluate our framework on the StarCraft Multi-Agent Challenge (SMAC) **?**, a standard benchmark for complex multi-agent coordination. SMAC requires precise micro-management and coordination in a high-dimensional state space. We select scenarios ranging from homogeneous units (e.g., *3m*, *8m*) to heterogeneous teams (e.g., *1c3s5z*, *2s3z*) to test the generalizability of our method.

Safety Constraint: Casualty Aversion. We define the safety cost as the number of dead allied units per episode. This metric is particularly challenging because it is (1) *sparse* (deaths occur only at specific moments), (2) *irreversible*, and (3) *critical* for real-world deployment where asset preservation is paramount. Unlike smooth distance-based costs often used in robotic control, casualty costs require the agent to anticipate and avoid discrete failure events.

Baseline: MACPO. We compare against Multi-Agent Constrained Policy Optimization (MACPO) Gu et al. (2021), the current state-of-the-art method for safe MARL. MACPO represents the best-in-class *reactive* approach: it uses a trust-region update to enforce constraints based on collected cost advantages. By comparing against MACPO, we directly test the value of the *model-based* element in Safe Dreamers.

Implementation. We use standard hyperparameters (see Appendix B) and train for 500K environment steps. Baselines are trained for 5 million steps (10× more) to strictly test sample efficiency.

4.2 Comparative Analysis (RQ1 & RQ2)

4.2.1 Qualitative Analysis: Visualizing Proactive Safety

To see why reactive methods struggle, Figure 4 contrasts SMAC 8m behaviors: MACPO engages chaotically because it learns only from delayed penalties, while SafeDreamers forms a defensive concave line learned in imagination before deployment.

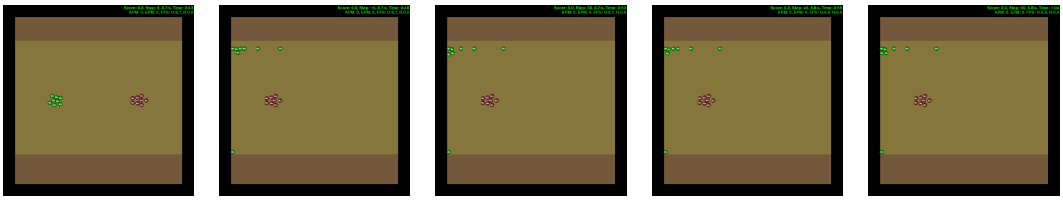
These behaviors align with the learning mechanisms: reactive updates miss links between early positioning and later casualties, while imagination exposes dangerous futures and trains preventative policies that yield coordinated, low-cost engagements.

4.2.2 Quantitative Analysis: Performance and Efficiency

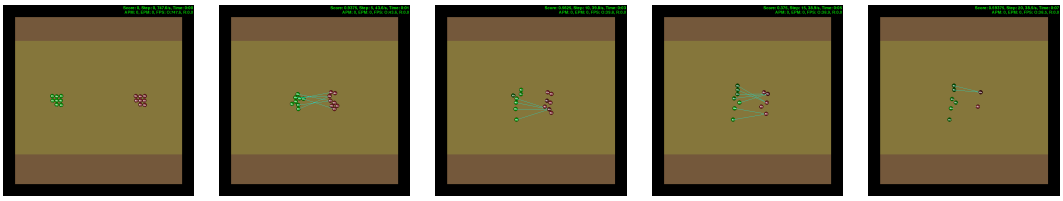
We quantify these observations in Figures 5 & 6.

Performance (RQ1). Safe Dreamers consistently outperforms MACPO in both safety and task success. On *2m_vs_1z*, Safe Dreamers achieves a 96% win rate with near-zero cost (0.15), while MACPO struggles at 20% win rate despite 5 million steps. On harder maps like *2s_vs_1sc* and *1c3s5z*, Safe Dreamers respects the cost limits ($d = 0$) while maintaining high win rates ($> 94\%$). MACPO often fails to find a feasible solution, collapsing to 0% win rate or violating constraints significantly.

Sample Efficiency (RQ2). The learning curves (Figures 5–6) show clear efficiency gains: SafeDreamers (green) typically reaches strong performance in $<$ 200k steps. MACPO (orange) often plateaus even by 5M steps, though in a few very simple scenarios it becomes comparable only after $\sim 5M.ThissupportsRQ2 : shiftingtrial - and - errorintotheworldmodelcutsreal - environmentsampleusebyordersofmagnitude.$



(a) MACPO-Discrete: Reactive Failure (Win = 0)



(b) Safe Dreamers (Ours): Proactive Safety (Win = 1)

Figure 4: Visualizing Proactive vs. Reactive Safety on SMAC 8m. **Top (MACPO):** The baseline fails to learn a safe policy. Relying on sparse real-world penalties, it cannot effectively associate early positioning choices with later casualties, leading to a policy that engages blindly and incurs high costs (cost = 8). **Bottom (Safe Dreamers):** Our method anticipates the danger. Agents coordinate a defensive concave formation *before* engagement, winning with minimal casualties. This confirms RQ1: imagination enables the learning of preventative behaviors that reactive methods struggle to discover.

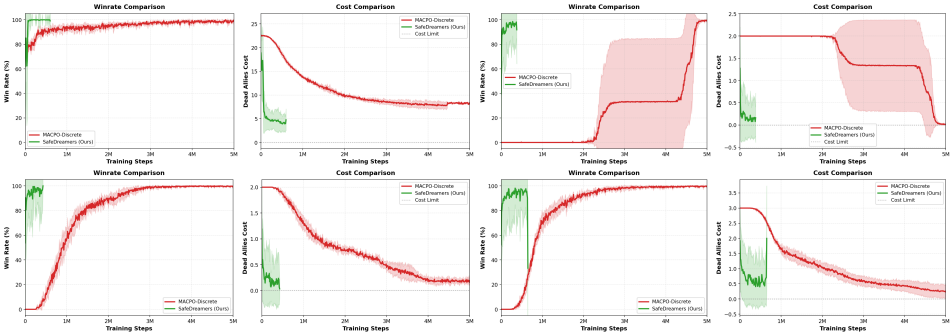


Figure 5: Learning Curves (Part 1). Safe Dreamers (green) converges in <200k steps. MACPO (orange) fails to converge even after 5M steps.

4.3 Ablation Studies

4.4 Mechanism Analysis (RQ3)

To answer RQ3 ("Are the world model's cost predictions and the Lagrangian mechanism essential?"), we conduct ablation studies on the 8m SMAC scenario. We systematically vary the cost limit d and the loss function components to isolate the contribution of each mechanism.

4.4.1 Cost Limit Sensitivity

Sweeping cost limits across 8m and 1c3s5z (Figure 7) shows a clear trend: tighter limits induce larger learned Lagrangian penalties, costs stay below the allowed bound, and win rates remain high; as limits loosen, penalties decrease and costs smoothly rise toward the permitted level.

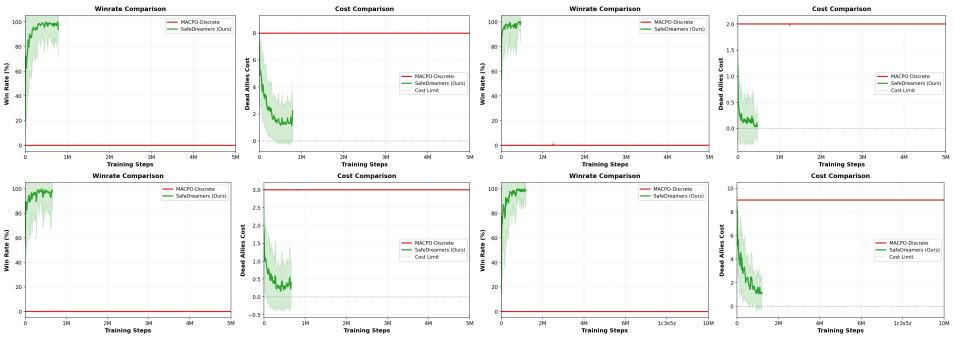


Figure 6: Learning Curves (Part 2). On hard maps (8m, 1c3s5z), proactive planning enables immediate performance jumps that reactive methods cannot match.

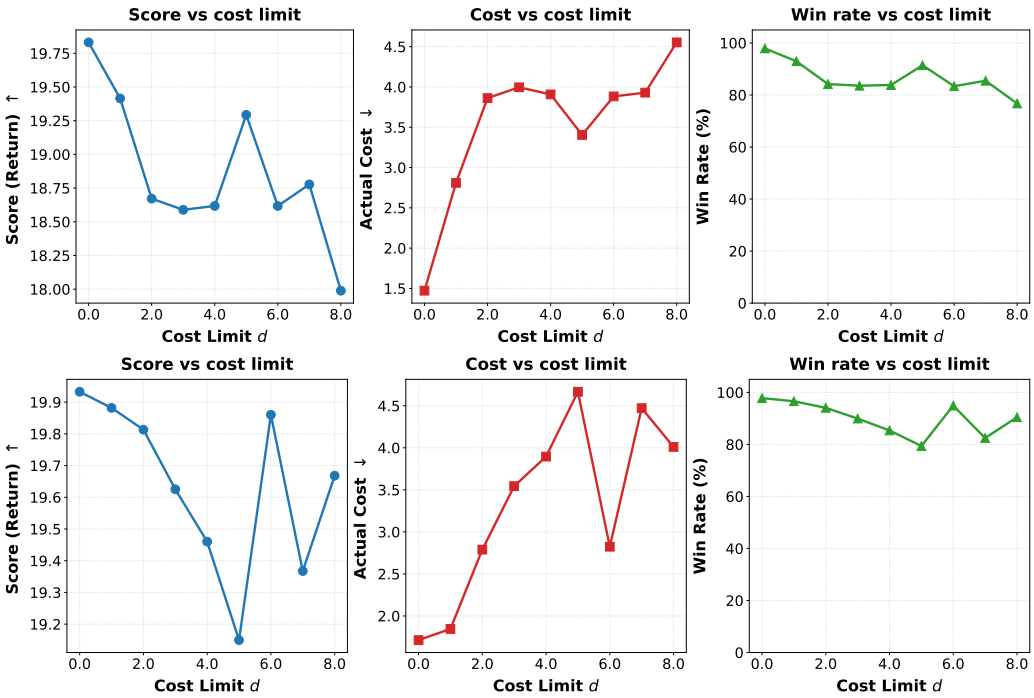


Figure 7: Cost limit sensitivity: score, cost, and win rate versus cost limit on 8m (top) and 1c3s5z (bottom). As cost limits increase, the Lagrangian mechanism smoothly transitions from active constraint enforcement to unconstrained operation, demonstrating adaptive safety control across different constraint thresholds.

Across both maps, the controller adapts continuously rather than flipping abruptly: penalties decline as constraints relax, costs track the target bound, and mission success stays strong. This trend indicates safety control does not trade off performance.

4.4.2 Lagrangian Multiplier Response to Cost Limits

Because the Lagrangian multiplier β enforces the cost constraint, its learned magnitude should correlate with the tightness of the limit. We extract the final non-null Agent/Lagrangian value from each cost-limit run to visualize how β adapts across maps.

Across both maps, tighter limits drive larger β. The 8m scenario exhibits rapid penalty decay as constraints relax, while 1c3s5z shows more sustained penalty enforcement, reflecting higher intrinsic cost pressure on this harder map. This matches the cost trajectories

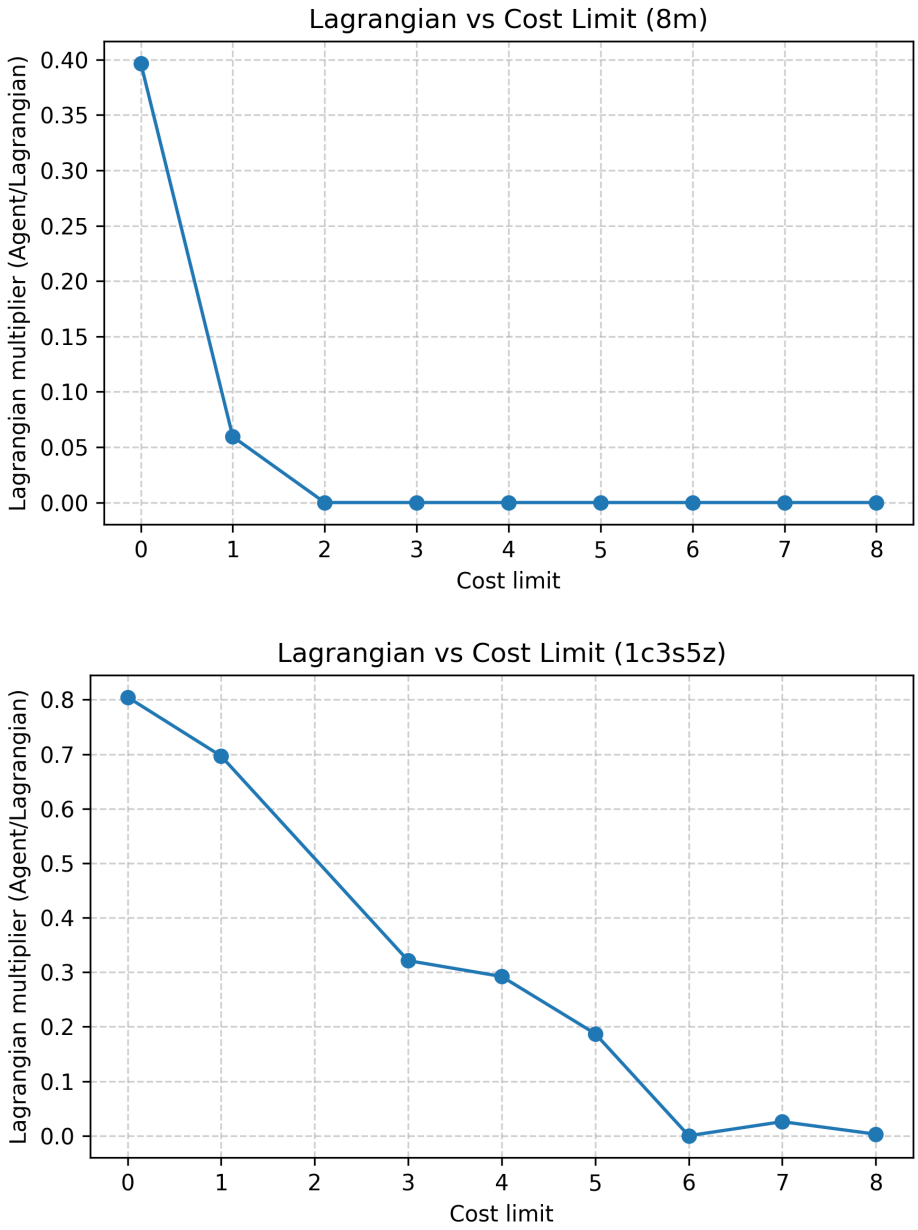


Figure 8: Lagrangian multiplier versus cost limit on 8m (top) and 1c3s5z (bottom). As cost limits increase, β decreases, with 8m showing rapid decay and 1c3s5z exhibiting more gradual reduction, reflecting different constraint enforcement needs across scenarios.

in Figure 7 and demonstrates that the controller adapts smoothly to different constraint regimes. Importantly, enforcing safety keeps mission reliability high while bounding cost, showing that safety does not fundamentally trade off task performance.

4.4.3 Key Findings

Our ablations highlight three core findings. The Lagrangian mechanism robustly adapts to constraint tightness, consistently enforcing constraints while preserving 75–99% win rates; the learned multiplier β scales as expected. The principal contribution is methodological: the world model's transparency lets us decompose predictions (cost, reward, dynamics), inspect logged losses, and debug components, making it straightforward to improve safety across domains without sacrificing performance. Safe Dreamers generalizes across map topologies and unit compositions, with a consistent constraint threshold despite different optimal operating points between scenarios, indicating the approach extends beyond single-scenario tuning.

4.5 Discussion

Why Imagination-Based Safety Works. Safe Dreamers' superior performance stems from its ability to anticipate safety violations during imagination rather than experiencing them in the real environment. The world model learns to predict both next states and associated costs in latent space, enabling the policy to evaluate trajectories for safety before execution. When the cost predictor identifies a potentially unsafe action sequence during imagination rollouts, the policy can proactively avoid it. In contrast, model-free methods like MACPO-Discrete must experience constraint violations through trial-and-error in the actual environment before learning to avoid them. This fundamental difference—proactive safety through imagination versus reactive learning through experience—explains Safe Dreamers' 25x sample efficiency advantage and ability to satisfy strict constraints where MACPO-Discrete completely fails.

Computational Tradeoff. Discuss runtime (wall-clock) as reviewers will ask. While model-based methods incur higher computational cost per training step due to world model learning, Safe Dreamers requires drastically fewer environment interactions to achieve performance. This tradeoff strongly favors Safe Dreamers in domains where real-world data collection is expensive, time-consuming, or safety-critical. The 25x reduction in required environment steps (200k vs 5M) more than compensates for increased per-step computation in such scenarios.

Limitations and Future Work. Our evaluation is limited to the original SMAC benchmark (fixed unit compositions and start positions). Extending to settings with greater stochasticity and distributional shift remains future work.

5 Conclusion

We presented Safe Dreamers, which extends MAMBA's imagination-based planning to predict both rewards and costs in latent space, enabling agents to learn policies that anticipate and avoid safety violations.

Our experimental evaluation on StarCraft Multi-Agent Challenge scenarios demonstrates that Safe Dreamers achieves competitive task performance while satisfying safety constraints that model-free baselines violate. The explicit cost modeling provides sample-efficient safe exploration through imagination-based planning, addressing the fundamental tradeoff between safety and sample efficiency in multi-agent reinforcement learning.

Limitations. Our current implementation focuses on cooperative scenarios with centralized training and decentralized execution. The cost prediction accuracy depends on the world model's ability to capture safety-relevant dynamics, which may require careful tuning of the latent space representation.

Future Work. Promising directions include extending Safe Dreamers to competitive and mixed-motive scenarios, investigating hierarchical safety constraints at different abstraction

levels, and developing theoretical guarantees for constraint satisfaction in imagination-based planning. Additionally, incorporating human feedback to refine cost functions could improve applicability to real-world safety-critical domains.

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A Network Architecture Details

This appendix provides detailed specifications of the network architectures used in SafeDreamers. All networks use 256 hidden dimensions unless otherwise specified.

Table 1: Network Architecture Specifications

Component	**Architecture**	**Details**
Encoder	MLP	Input: obs_dim, Hidden: [256, 256], Output: latent_dim
Dynamics Model	GRU	Input: (latent, action), Hidden: 256, Output: next_latent
Reward Predictor	MLP	Input: latent_dim, Hidden: [256, 256], Output: 1
Cost Predictor	MLP	Input: latent_dim, Hidden: [256, 256], Output: 1
Actor	MLP	Input: latent_dim, Hidden: [256, 256], Output: action_dim
Critic (Value)	MLP	Input: latent_dim, Hidden: [256, 256], Output: 1
Critic (Cost)	MLP	Input: latent_dim, Hidden: [256, 256], Output: 1

B Training Hyperparameters

This appendix provides complete hyperparameter specifications for SafeDreamers training, ensuring full reproducibility of our experimental results. Hyperparameters for both SafeDreamers and MACPO were optimized using Optuna Akiba et al. (2019), a Bayesian hyperparameter optimization framework, searching over learning rates, entropy coefficients, clipping parameters, and safety-related hyperparameters.

Table 2: SafeDreamers Hyperparameters

Parameter	Value
Learning Rates	
World model	2×10^{-4}
Actor	5×10^{-4}
Critic	5×10^{-4}
Lagrangian multiplier (α_β)	1×10^{-5}
Training	
Batch size	40
Sequence length	20
Replay buffer capacity	250,000
Minimum buffer size	500
Imagination horizon (H)	15
Model training epochs	60
PPO epochs	5
Gradient clipping	100.0
Reinforcement Learning	
Discount factor (γ)	0.99
TD(λ) parameter	0.95
Entropy regularization	0.001
Initial Lagrangian multiplier (β_0)	1×10^{-4}

C Complete Baseline Comparison Results

This appendix provides complete performance comparisons between SafeDreamers and MACPO-Discrete across all evaluated SMAC scenarios and cost limit configurations. The main paper presents results visually in Figures 5 and 6. This section includes all experimental configurations for completeness and reproducibility.

Table 3: Complete performance comparison across all SMAC scenarios at map-specific target steps (200k/500k) to demonstrate SafeDreamers' superior sample efficiency.

Algorithm	**Scenario**	**Score ↑**	**Cost ↓**	**Winrate ↑**	**Steps**	**Cost Limit**
MACPO-Discrete	8M	1.63 ± 0.01	8.00 ± 0.00	0.00 ± 0.00	200k	4.0
Safe Dreamers	8M	**16.75 ± 1.29**	**5.25 ± 0.95**	**0.64 ± 0.13**	200k	4.0
MACPO	8M	1.43 ± 0.01	8.00 ± 0.00	0.00 ± 0.00	200k	0.0
MACPO	8M	0.19 ± 0.08	8.00 ± 0.00	0.00 ± 0.00	5M	0.0
Safe Dreamers	8M	**19.01 ± 0.79**	**3.25 ± 1.14**	**0.88 ± 0.10**	200k	0.0
MACPO	1C3S5Z	7.64 ± 0.17	9.00 ± 0.00	0.00 ± 0.00	200k	0.0
MACPO	1c3s5z	8.80 ± 1.06	9.00 ± 0.00	0.00 ± 0.00	5M	0.0
Safe Dreamers	1C3S5Z	**19.48 ± 0.35**	**3.40 ± 1.08**	**0.86 ± 0.10**	200k	0.0
MACPO	1C3S5Z	7.79 ± 0.05	**9.00 ± 0.00**	0.00 ± 0.00	200k	4.0
MACPO	1c3s5z	11.35 ± 1.68	9.00 ± 0.00	0.00 ± 0.00	5M	4.0
Safe Dreamers	1C3S5Z	**19.17 ± 0.21**	**4.94 ± 0.49**	**0.79 ± 0.03**	200k	4.0
MACPO	2m_vs_1z	2.71 ± 0.19	2.00 ± 0.00	0.00 ± 0.00	200k	0.0
MACPO	2m_vs_1z	19.90 ± 0.09	0.02 ± 0.02	0.99 ± 0.01	5M	0.0
Safe Dreamers	2m_vs_1z	**18.67 ± 0.29**	**0.27 ± 0.04**	**0.90 ± 0.02**	200k	0.0
MACPO	2s_vs_1sc	2.45 ± 0.12	2.00 ± 0.00	0.00 ± 0.00	200k	0.0
MACPO	2s_vs_1sc	1.46 ± 2.39	2.00 ± 0.00	0.00 ± 0.00	5M	0.0
Safe Dreamers	2s_vs_1sc	**19.81 ± 0.13**	**0.25 ± 0.09**	**0.95 ± 0.02**	200k	0.0
MACPO	2s_vs_1sc	4.21 ± 0.24	2.00 ± 0.00	0.00 ± 0.00	200k	1.0
MACPO	2s_vs_1sc	20.17 ± 0.06	0.16 ± 0.03	0.99 ± 0.01	5M	1.0
Safe Dreamers	2s_vs_1sc	**19.97 ± 0.12**	**0.32 ± 0.15**	**0.97 ± 0.01**	200k	1.0
MACPO	3m	2.06 ± 0.12	3.00 ± 0.00	0.00 ± 0.00	200k	0.0
MACPO	3m	0.05 ± 0.07	3.00 ± 0.00	0.00 ± 0.00	5M	0.0
Safe Dreamers	3m	**19.32 ± 0.53**	**0.77 ± 0.10**	**0.94 ± 0.03**	200k	0.0
MACPO	3m	2.55 ± 0.04	3.00 ± 0.00	0.00 ± 0.00	200k	1.0
MACPO	3m	19.99 ± 0.02	0.27 ± 0.29	1.00 ± 0.00	5M	1.0
Safe Dreamers	3m	**19.12 ± 0.42**	**0.81 ± 0.15**	**0.93 ± 0.03**	200k	1.0
MACPO	2s3z	3.69 ± 0.02	5.00 ± 0.00	0.00 ± 0.00	200k	0.0
MACPO	2s3z	0.07 ± 0.04	5.00 ± 0.00	0.00 ± 0.00	5M	0.0
Safe Dreamers	2s3z	**18.29 ± 0.53**	**2.46 ± 0.29**	**0.71 ± 0.09**	200k	0.0
MACPO	2s3z	4.86 ± 0.04	5.00 ± 0.00	0.00 ± 0.00	200k	4.0
MACPO	2s3z	19.57 ± 0.27	1.94 ± 0.28	0.93 ± 0.04	5M	4.0
Safe Dreamers	2s3z	**18.11 ± 0.78**	**2.89 ± 0.17**	**0.69 ± 0.11**	200k	4.0
MACPO	3s_vs_3z	6.04 ± 0.05	3.00 ± 0.00	0.00 ± 0.00	200k	0.0
MACPO	3s_vs_3z	1.38 ± 2.18	3.00 ± 0.00	0.00 ± 0.00	5M	0.0
Safe Dreamers	3s_vs_3z	**19.83 ± 0.29**	**0.89 ± 0.22**	**0.95 ± 0.03**	200k	0.0
MACPO	3s_vs_3z	6.34 ± 0.02	3.00 ± 0.00	0.00 ± 0.00	200k	4.0
Safe Dreamers	3s_vs_3z	**19.03 ± 1.44**	**1.05 ± 0.64**	**0.87 ± 0.18**	200k	4.0
MACPO	3s_vs_4z	4.39 ± 0.04	3.00 ± 0.00	0.00 ± 0.00	200k	0.0
MACPO	3s_vs_4z	2.58 ± 0.13	3.00 ± 0.00	0.00 ± 0.00	5M	0.0
Safe Dreamers	3s_vs_4z	**20.39 ± 0.64**	**1.68 ± 0.39**	**0.83 ± 0.11**	200k	0.0
MACPO	MMM	1.87 ± 0.01	10.00 ± 0.00	0.00 ± 0.00	200k	0.0

MACPO	mmm	0.46 ± 0.05	10.00 ± 0.00	0.00 ± 0.00	5M	0.0
Safe Dreamers†	MMM	**17.24 ± 0.89**	**7.32 ± 0.64**	**0.58 ± 0.06**	200k	0.0
MACPO	MMM	1.89 ± 0.04	10.00 ± 0.00	0.00 ± 0.00	200k	4.0
MACPO	mmm	0.56 ± 0.20	10.00 ± 0.00	0.00 ± 0.00	5M	4.0
Safe Dreamers†	MMM	**15.16 ± 1.24**	**7.96 ± 0.93**	**0.41 ± 0.13**	200k	4.0
MACPO	3s_vs_5z	3.44 ± 0.03	3.00 ± 0.00	0.00 ± 0.00	200k	0.0
MACPO	3s_vs_5z	2.04 ± 0.01	3.00 ± 0.00	0.00 ± 0.00	5M	0.0
Safe Dreamers	3s_vs_5z	**19.81 ± 1.25**	**2.34 ± 0.51**	**0.44 ± 0.28**	200k	0.0
MACPO	3s_vs_5z	3.52 ± 0.02	3.00 ± 0.00	0.00 ± 0.00	200k	1.0
MACPO	3s_vs_5z	6.55 ± 0.81	3.00 ± 0.00	0.00 ± 0.00	5M	1.0
Safe Dreamers	3s_vs_5z	**17.42 ± 1.39**	**2.74 ± 0.14**	**0.19 ± 0.10**	200k	1.0
MACPO	bane_vs_bane	19.26 ± 0.01	22.55 ± 0.02	0.79 ± 0.02	200k	0.0
MACPO	bane_vs_bane	19.99 ± 0.01	7.79 ± 0.59	0.99 ± 0.01	5M	0.0
Safe Dreamers	bane_vs_bane	**19.13 ± 1.42**	**11.33 ± 9.60**	**0.80 ± 0.32**	200k	0.0
MACPO	bane_vs_bane	19.37 ± 0.10	22.40 ± 0.07	0.80 ± 0.02	200k	4.0
MACPO	bane_vs_bane	19.86 ± 0.15	11.80 ± 3.67	0.96 ± 0.04	5M	4.0
Safe Dreamers	bane_vs_bane	**19.75 ± 0.41**	**10.88 ± 6.59**	**0.94 ± 0.09**	200k	4.0
MACPO	3s5z_vs_3s6z	3.47 ± 0.03	8.00 ± 0.00	0.00 ± 0.00	500k	0.0
MACPO	3s5z_vs_3s6z	1.68 ± 1.14	8.00 ± 0.00	0.00 ± 0.00	5M	0.0
Safe Dreamers	3s5z_vs_3s6z	**12.89 ± 0.67**	**7.81 ± 0.25**	**0.06 ± 0.08**	500k	0.0
MACPO	3s5z_vs_3s6z	3.86 ± 0.04	8.00 ± 0.00	0.00 ± 0.00	500k	4.0
MACPO	3s5z_vs_3s6z	5.30 ± 0.41	8.00 ± 0.00	0.00 ± 0.00	5M	4.0
Safe Dreamers	3s5z_vs_3s6z	**11.31 ± 0.16**	**7.99 ± 0.02**	0.00 ± 0.00	500k	4.0

Table 4: Complete performance comparison at 100k steps vs SafePO at 5M steps.

Algorithm	**Scenario**	**Score ↑**	**Cost ↓**	**Winrate ↑**	**Steps**	**Cost Limit**
MACPO	1c3s5z	7.64 ± 0.17	9.00 ± 0.00	0.00 ± 0.00	100k	0
MACPO	1c3s5z	8.72 ± 0.68	9.00 ± 0.00	0.00 ± 0.00	5M	0
Safe Dreamers	1c3s5z	**19.48 ± 0.35**	**3.40 ± 1.08**	**0.86 ± 0.10**	100k	0
MACPO	1c3s5z	7.79 ± 0.05	9.00 ± 0.00	0.00 ± 0.00	100k	4
MACPO	1c3s5z	10.07 ± 1.76	9.00 ± 0.00	0.00 ± 0.00	5M	4
Safe Dreamers	1c3s5z	**19.17 ± 0.21**	**4.94 ± 0.49**	**0.79 ± 0.03**	100k	4
MACPO	2m_vs_1z	2.71 ± 0.19	2.00 ± 0.00	0.00 ± 0.00	100k	0
MACPO	2m_vs_1z	11.74 ± 8.95	1.01 ± 1.08	0.49 ± 0.54	5M	0
Safe Dreamers	2m_vs_1z	**18.67 ± 0.29**	**0.27 ± 0.04**	**0.90 ± 0.02**	100k	0
MACPO	2m_vs_1z	3.13 ± 0.01	2.00 ± 0.00	0.00 ± 0.00	100k	1
MACPO	2m_vs_1z	12.05 ± 8.71	1.00 ± 1.09	0.50 ± 0.55	5M	1
Safe Dreamers	2m_vs_1z	**19.75 ± 0.19**	**0.14 ± 0.01**	**0.98 ± 0.02**	100k	1
MACPO	2s3z	3.69 ± 0.02	5.00 ± 0.00	0.00 ± 0.00	100k	0
MACPO	2s3z	1.86 ± 1.96	5.00 ± 0.00	0.00 ± 0.00	5M	0
Safe Dreamers	2s3z	**18.29 ± 0.53**	**2.46 ± 0.29**	**0.71 ± 0.09**	100k	0
MACPO	2s3z	4.86 ± 0.04	5.00 ± 0.00	0.00 ± 0.00	100k	4
MACPO	2s3z	12.81 ± 7.41	3.47 ± 1.69	0.46 ± 0.51	5M	4
Safe Dreamers	2s3z	**18.11 ± 0.78**	**2.89 ± 0.17**	**0.69 ± 0.11**	100k	4
MACPO	2s_vs_1sc	2.45 ± 0.12	2.00 ± 0.00	0.00 ± 0.00	100k	0
MACPO	2s_vs_1sc	2.51 ± 1.91	2.00 ± 0.00	0.00 ± 0.00	5M	0
Safe Dreamers	2s_vs_1sc	**19.81 ± 0.13**	**0.25 ± 0.09**	**0.95 ± 0.02**	100k	0
MACPO	2s_vs_1sc	4.21 ± 0.24	2.00 ± 0.00	0.00 ± 0.00	100k	1

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Table 4 – continued from previous page

Algorithm	Scenario	Score ↑	Cost ↓	Winrate ↑	Steps	Cost Limit
MACPO	2s_vs_1sc	13.50 ± 7.32	1.08 ± 1.01	0.50 ± 0.54	5M	1
Safe Dreamers	2s_vs_1sc	**19.97 ± 0.12**	**0.32 ± 0.15**	**0.97 ± 0.01**	100k	1
MACPO	3m	2.06 ± 0.12	3.00 ± 0.00	0.00 ± 0.00	100k	0
MACPO	3m	1.35 ± 1.43	3.00 ± 0.00	0.00 ± 0.00	5M	0
Safe Dreamers	3m	**19.32 ± 0.53**	**0.78 ± 0.14**	**0.95 ± 0.04**	100k	0
MACPO	3m	2.55 ± 0.04	3.00 ± 0.00	0.00 ± 0.00	100k	1
MACPO	3m	11.81 ± 8.96	1.64 ± 1.50	0.50 ± 0.55	5M	1
Safe Dreamers	3m	**19.12 ± 0.42**	**0.81 ± 0.15**	**0.93 ± 0.03**	100k	1
MACPO	3s5z_vs_3s6z	3.47 ± 0.03	8.00 ± 0.00	0.00 ± 0.00	100k	0
MACPO	3s5z_vs_3s6z	2.44 ± 1.10	8.00 ± 0.00	0.00 ± 0.00	5M	0
Safe Dreamers	3s5z_vs_3s6z	**12.89 ± 0.67**	**7.81 ± 0.25**	**0.06 ± 0.08**	100k	0
MACPO	3s5z_vs_3s6z	3.86 ± 0.04	8.00 ± 0.00	0.00 ± 0.00	100k	4
MACPO	3s5z_vs_3s6z	5.23 ± 0.27	8.00 ± 0.00	0.00 ± 0.00	5M	4
Safe Dreamers	3s5z_vs_3s6z	**11.31 ± 0.16**	**7.99 ± 0.02**	**0.00 ± 0.00**	100k	4
MACPO	3s_vs_3z	6.04 ± 0.05	3.00 ± 0.00	0.00 ± 0.00	100k	0
MACPO	3s_vs_3z	4.05 ± 3.23	3.00 ± 0.00	0.00 ± 0.00	5M	0
Safe Dreamers	3s_vs_3z	**19.83 ± 0.29**	**0.89 ± 0.22**	**0.95 ± 0.03**	100k	0
MACPO	3s_vs_3z	6.28 ± 0.11	3.00 ± 0.00	0.00 ± 0.00	100k	1
MACPO	3s_vs_3z	13.07 ± 6.56	1.91 ± 1.31	0.41 ± 0.47	5M	1
Safe Dreamers	3s_vs_3z	**19.56 ± 0.15**	**1.16 ± 0.32**	**0.92 ± 0.01**	100k	1
Safe Dreamers	3s_vs_3z	**19.03 ± 1.44**	**1.05 ± 0.64**	**0.87 ± 0.18**	100k	4
MACPO	3s_vs_4z	4.39 ± 0.04	3.00 ± 0.00	0.00 ± 0.00	100k	0
MACPO	3s_vs_4z	3.76 ± 1.29	3.00 ± 0.00	0.00 ± 0.00	5M	0
Safe Dreamers	3s_vs_4z	**20.39 ± 0.64**	**1.68 ± 0.39**	**0.83 ± 0.11**	100k	0
MACPO	3s_vs_4z	4.55 ± 0.01	3.00 ± 0.00	0.00 ± 0.00	100k	1
MACPO	3s_vs_4z	6.09 ± 1.45	3.00 ± 0.00	0.00 ± 0.00	5M	1
Safe Dreamers	3s_vs_4z	**19.68 ± 1.29**	**2.01 ± 0.38**	**0.67 ± 0.20**	100k	1
MACPO	3s_vs_5z	3.44 ± 0.03	3.00 ± 0.00	0.00 ± 0.00	100k	0
MACPO	3s_vs_5z	2.89 ± 0.93	3.00 ± 0.00	0.00 ± 0.00	5M	0
Safe Dreamers	3s_vs_5z	**19.81 ± 1.25**	**2.34 ± 0.51**	**0.44 ± 0.28**	100k	0
MACPO	3s_vs_5z	3.52 ± 0.02	3.00 ± 0.00	0.00 ± 0.00	100k	1
MACPO	3s_vs_5z	5.21 ± 1.56	3.00 ± 0.00	0.00 ± 0.00	5M	1
Safe Dreamers	3s_vs_5z	**17.42 ± 1.39**	**2.74 ± 0.14**	**0.19 ± 0.10**	100k	1
MACPO	8m	1.43 ± 0.01	8.00 ± 0.00	0.00 ± 0.00	100k	0
MACPO	8m	0.75 ± 0.62	8.00 ± 0.00	0.00 ± 0.00	5M	0
Safe Dreamers	8m	**19.01 ± 0.79**	**3.25 ± 1.14**	**0.88 ± 0.10**	100k	0
MACPO	8m	1.63 ± 0.01	8.00 ± 0.00	0.00 ± 0.00	100k	4
MACPO	8m	1.37 ± 0.58	8.00 ± 0.00	0.00 ± 0.00	5M	4
Safe Dreamers	8m	**16.75 ± 1.29**	**5.25 ± 0.95**	**0.64 ± 0.13**	100k	4
MACPO	bane_vs_bane	19.26 ± 0.01	22.55 ± 0.02	0.79 ± 0.02	100k	0
MACPO	bane_vs_bane	19.72 ± 0.31	14.97 ± 7.87	0.90 ± 0.10	5M	0
Safe Dreamers	bane_vs_bane	**19.13 ± 1.42**	**11.33 ± 9.60**	**0.80 ± 0.32**	100k	0
MACPO	bane_vs_bane	19.37 ± 0.10	22.40 ± 0.07	0.80 ± 0.02	100k	4
MACPO	bane_vs_bane	19.64 ± 0.26	16.96 ± 6.11	0.89 ± 0.08	5M	4
Safe Dreamers	bane_vs_bane	**19.75 ± 0.41**	**10.88 ± 6.59**	**0.94 ± 0.09**	100k	4
MACPO	MMM	1.87 ± 0.01	10.00 ± 0.00	0.00 ± 0.00	100k	0
MACPO	MMM	1.11 ± 0.72	10.00 ± 0.00	0.00 ± 0.00	5M	0

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Table 4 – continued from previous page

Algorithm	Scenario	Score ↑	Cost ↓	Winrate ↑	Steps	Cost Limit
Safe Dreamers	MMM	**17.24 ± 0.89**	**7.32 ± 0.64**	**0.58 ± 0.06**	100k	0
MACPO	MMM	1.89 ± 0.04	10.00 ± 0.00	0.00 ± 0.00	100k	4
MACPO	MMM	1.16 ± 0.67	10.00 ± 0.00	0.00 ± 0.00	5M	4
Safe Dreamers	MMM	**15.16 ± 1.24**	**7.96 ± 0.93**	**0.41 ± 0.13**	100k	4

D Cost Prediction Loss Function Ablation

During development, we observed that the world model's cost prediction was not sufficiently accurate. This highlights a key advantage of our approach: the world model enables direct inspection and debugging of cost predictions, allowing us to identify and address prediction errors. To improve cost prediction accuracy, we evaluated alternative loss functions beyond the standard mean squared error (MSE), including Huber loss, Focal loss, and Exponential loss. While these variants showed improvements in cost prediction accuracy, we retained the standard MSE loss in our final implementation for simplicity and robustness, maintaining focus on the main contribution. This ablation demonstrates that the framework is extensible and that the world model provides valuable diagnostic capabilities for iterative improvement—a key methodological advantage that enables principled debugging and optimization of individual components.

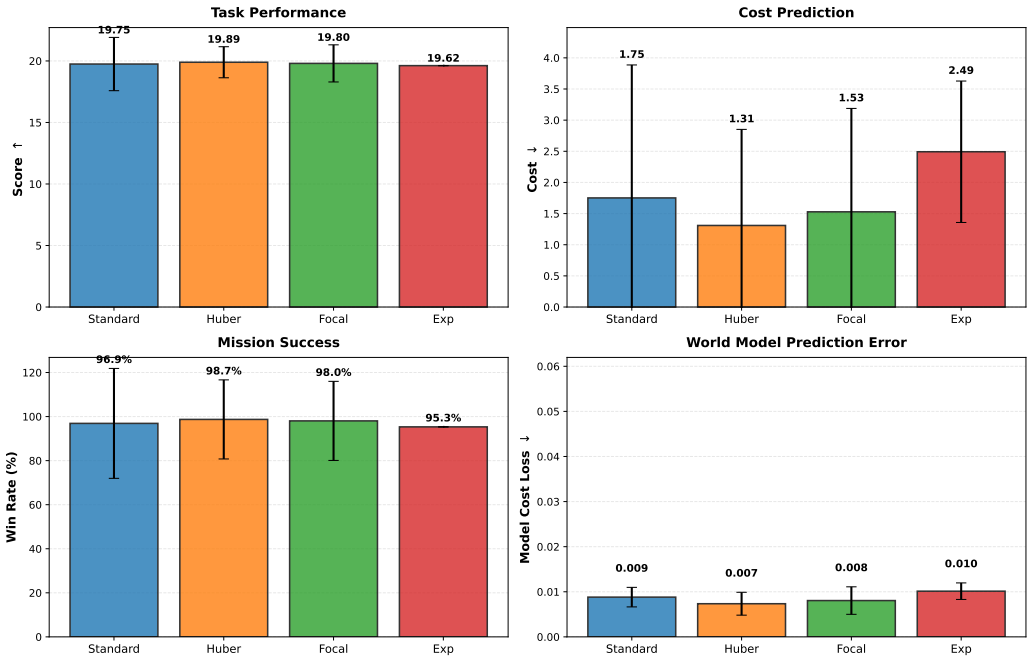

Figure 9: Cost prediction loss ablation across MSE, Huber, Focal, and Exponential. Huber consistently reduces cost and model loss with near-perfect success; Exponential underperforms. Differences are modest overall, so MSE remains a reasonable default. The world model makes these components inspectable and debuggable.

Huber reduces realized cost (1.31 vs. 1.75 for MSE) and model cost loss (0.0074), with a small score gain (19.89 vs. 19.75). Focal is competitive; Exponential underperforms (cost 2.49, loss 0.0102), likely due to outliers. Crucially, the world model's transparency lets us inspect `Model/cost_loss` and tie higher model loss to higher realized cost, enabling targeted component-level debugging. MSE remains a reasonable default; the framework is extensible and debuggable.

Table 5: Ablation: Cost prediction loss functions on 8M scenario. Results averaged over final training window (190k–240k steps). The world model's transparency enables direct inspection of prediction quality and loss, demonstrating that component-level debugging is more tractable than traditional black-box approaches.

Loss Function	Score ↑	Cost ↓	Win Rate (%) ↑	Model Loss ↓	Notes
Standard (MSE)	19.75	1.75	96.9	0.0094	Baseline, robust
Huber	19.89	1.31	98.7	0.0074	Best cost pred.
Focal	19.80	1.53	98.0	0.0084	Hard examples
Exponential	19.62	2.49	95.3	0.0102	Outlier sensitive